

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 2.2 PLANT TRANSPORT: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use everything you learned in Lessons 1 to 8 to answer the driving question: How does water and food travel through an entire plant, from the soil to the highest leaf, and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and die? Remember the Kibwezi farmer: she watered her maize in the morning, yet it wilted in the afternoon. Maize near the dry seasonal riverbed wilted sooner than maize on the wetter lower slopes, and sukuma wiki shaded by a mango tree barely wilted at all. Answer each section in full sentences in the space under each prompt. Use the key words you learned, such as xylem, phloem, osmosis, active transport, transpiration, cohesion-tension, turgor and pressure-flow. Where a section asks for a Claim-Evidence-Reasoning (CER) answer, give your claim, then your evidence from our investigations, then your reasoning. You may add a labelled diagram if it helps. Write in your own words.

Section 1: Why a plant needs a transport system

The Kibwezi farmer says, 'The soil has water, I can see it, so why is my maize wilting?' (a) Where do water and mineral salts enter a maize plant, and where does carbon dioxide enter? (b) Where in the plant is the water needed? (c) Use your answers to (a) and (b) to explain the farmer's puzzle. Name the two transport tissues that connect the entry points to the places of use.

Write your answer here:

Section 2: Entry of water and mineral salts at the root

Describe the path taken by water from the soil to the xylem in a maize root, naming the structures it passes through. (a) What process moves the water, and what makes it move in that direction? (b) How do mineral salts enter the root, and how is this different from water? (c) In your potato strip investigation, you used tap water and a concentrated salt solution. Each potato strip started 50 mm long. After 15 minutes the strip in tap water measured 53 mm and the strip in concentrated salt solution measured 46 mm.

Solution	Start length (mm)	Length after 15 min (mm)
Tap water	50	53
Concentrated salt solution	50	46

Use these results to explain what the potato strips modelled for the root, and how this can help explain why maize in drier soil near the seasonal riverbed wilts sooner than maize on the wetter lower slopes.

Write your answer here:

Section 3: The journey up the stem and the loss at the leaf

Using Claim-Evidence-Reasoning, answer: Why does the Kibwezi maize wilt in the afternoon even though the farmer watered the soil that morning? In your Evidence, use what you observed in our leaf investigations and this class potometer data (a potometer measures how fast a cut leafy stem takes up water; the air bubble moved along the capillary tube for 2 minutes in each condition).

Condition	Bubble distance in 2 min (mm)	Uptake rate (mm/min)
Control (still air)	12	6
Wind (fan)	30	15
Heat (lamp)	26	13
Petroleum jelly on leaves	4	2

In your Reasoning, explain how water is pulled up from the roots to the leaves, and link the conditions that change the rate of water loss to a Kibwezi afternoon.

Write your answer here:

Section 4: Why some plants recover, some die, and shade protects sukuma wiki

The farmer noticed that by evening some maize plants recovered while others did not, and that the sukuma wiki under the mango tree barely wilted. (a) Explain why shade protects the sukuma wiki, naming which transpiration factors the shade changes. (b) Explain why some maize plants never recover, even after the farmer waters the soil again. Think about what could happen inside the stem when water loss is very high and supply is low. (c) Why do the shaded sukuma wiki plants have a lower risk of this problem?

Write your answer here:

Section 5: The return journey of food and the complete model

Plants also move food. (a) Which tissue carries sucrose and amino acids, from where to where, and what did the girdled eucalyptus trees near Kericho show you about this? (b) Explain the pressure-flow hypothesis in your own words. (c) Finally, summarise the whole plant transport journey in order, from water entering a root hair to sugar reaching a sink such as the maize cob, and say how this journey can fail in the Kibwezi maize.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of the transport pathways	Correctly describes water and mineral entry (osmosis, active transport), upward flow in xylem and sugar flow in phloem, using the correct terms throughout.	Describes most pathways correctly with minor errors or missing terms (for example confuses one direction or omits the endodermis).	Confuses xylem and phloem, or the processes, or gives incomplete pathways.
Use of evidence from the investigations	Cites specific evidence and numbers from our lessons (potato strips, sweating leaf bag, stomatal impressions, potometer, girdled eucalyptus trees near Kericho) and links each piece to a claim.	Cites some evidence from the investigations but links it to claims only partly.	Gives little or no evidence, or only general statements.
Explaining the Kibwezi phenomenon	Fully explains why maize wilts despite watered soil, why riverbed maize wilts sooner, why some plants die and why shaded sukuma wiki is protected, using rate of loss versus supply.	Explains most parts of the phenomenon but misses one (for example the shade or the riverbed explanation).	Explains only that the plant needs water without linking to transpiration or soil water potential.

Reasoning with mechanisms (CER, cohesion-tension, pressure-flow)	Gives clear, logical reasoning connecting cause and effect, including cohesion-tension, the four transpiration factors, and pressure-flow.	Gives reasoning that is mostly logical but with gaps in one mechanism.	States facts without explaining how or why.
Communication and Kenyan context	Writes clearly and in order, using the key terms correctly and linking ideas to the Kibwezi farm context.	Writes understandably with a few unclear parts or terms used loosely.	Writing is hard to follow and key terms are missing or misused.

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